

Position sizing dominates strategy selection

10,000 simulated equity curves on what determines whether a strategy enriches or ruins its trader

ABSTRACT

We construct 10,000 Monte Carlo equity-curve simulations across a parameterised family of trend-following systems, varying the position-sizing rule while holding the underlying signal generation constant. Five sizing approaches are tested: fixed-dollar, fixed-fractional, volatility-targeted, full Kelly, and fractional Kelly. Across the simulation universe, position sizing explains approximately 60% of the cross-sectional variance in terminal wealth and approximately 80% of the variance in maximum drawdown — substantially more than the variance attributable to signal-strength differences across reasonable parameter ranges. Two structurally different systems with identical signal Sharpe ratios but different position-sizing rules can produce wealth distributions that differ by an order of magnitude, with the better-sized system producing positive long-run growth and the poorer-sized system producing ruin in 30%+ of paths. We document the relative merits and failure modes of each sizing approach, examine the relationship between win rate and optimal sizing, and conclude with practical implications for system design and disclosure to end users.

KEY FINDINGS

- 01 Position sizing explains the majority of outcome variance.** Across 10,000 simulated paths, position sizing explains ~60% of variance in terminal wealth and ~80% of variance in maximum drawdown.
- 02 Identical signals can produce ruin or wealth.** Two strategies with identical signal Sharpe ratios but different sizing rules produce terminal wealth distributions that differ by 10x or more, with poor sizing producing ruin in 30%+ of paths.
- 03 Win rate is misleading.** A 70% win-rate system at the wrong size goes bankrupt in 25% of paths. A 55% win-rate system at the right size compounds steadily. Win rate is one of the least informative summary statistics.
- 04 Full Kelly is theoretically optimal but practically dangerous.** Full Kelly sizing produces highest geometric mean returns in expectation but also produces 50%+ drawdowns in roughly half of all paths. Practitioners use fractional Kelly (typically 0.25–0.5×) for behavioural and parameter-uncertainty reasons.

- 05 Volatility targeting dominates simpler sizing rules.** Volatility-targeted sizing (sizing positions to constant ex-ante portfolio volatility) produces both higher Sharpe and lower maximum drawdown than fixed-fractional or fixed-dollar sizing across most parameter combinations tested.

EXEC Executive summary

The trading literature spends most of its attention on signal generation — what to buy, what to sell, when, why. Position sizing — how much — receives perhaps a tenth of that attention. The empirical evidence suggests this allocation of attention is roughly inverted from what produces good outcomes.

This note constructs 10,000 Monte Carlo simulations of equity curves produced by trend-following systems with varying position-sizing rules but otherwise identical signal characteristics. The purpose is to isolate the contribution of sizing to terminal outcomes — a quantity that is intuitively obvious to experienced traders but rarely measured in the literature. The result is striking: across the simulation universe, position sizing explains approximately 60% of the cross-sectional variance in terminal wealth and approximately 80% of the variance in maximum drawdown.

The methodology proceeds as follows. We define a base strategy — a 12-month trend-following system on a multi-asset basket — with parameters that produce, in expectation, an annualised Sharpe ratio of approximately 0.7 across the simulation period. We then vary the position-sizing rule across five canonical approaches: fixed-dollar (constant notional per trade), fixed-fractional (constant percentage of equity per trade), volatility-targeted (scaled to constant ex-ante portfolio volatility), full Kelly (mathematically optimal sizing assuming known parameters), and fractional Kelly (typically 0.25–0.5 of full Kelly, the practitioner consensus). For each sizing rule, we simulate 2,000 forty-year equity-curve paths using bootstrapped monthly returns from the historical sample, producing 10,000 paths in total.

The cross-sectional results are sharp. Volatility-targeted sizing produces the most favourable distribution: median terminal wealth approximately 5× initial, 25th-percentile wealth approximately 2.5×, and only 4% of paths ending in 50%+ drawdown. Fractional Kelly is a close second. Fixed-fractional and fixed-dollar sizing produce wider distributions: higher upper tail but materially worse lower tail, with 18–24% of paths ending in 50%+ drawdown depending on the parameter calibration. Full Kelly produces the highest expected geometric mean but also produces 50%+ drawdowns in roughly half of all paths — a profile that is theoretically optimal but practically intolerable for most investors.

The decomposition of variance is informative. Within any single sizing approach, varying the signal-generation parameters across reasonable ranges (lookback windows of 6–18 months, threshold parameters within typical bounds) explains approximately 20% of cross-path variance. Varying the sizing approach itself explains approximately 60%. Bootstrap noise (the randomness in the path generation given a fixed signal and sizing rule) explains the remaining 20%. Position sizing therefore dominates both signal selection and pure luck as a determinant of outcome.

The implication for system design and disclosure is direct. A system marketed on its signal characteristics — Sharpe ratio, win rate, average gain — without explicit attention to position sizing is being marketed on the dimension that explains the smaller fraction of outcome variance. Users selecting between systems primarily on signal characteristics, rather than on the joint sizing-and-signal package, are optimising the wrong variable.

01 Introduction

The proposition that position sizing matters more than signal selection is well-established among experienced traders and futures-market practitioners but is under-represented in the academic and retail literature. [Vince \(1990, 1992\)](#) developed the formal framework for "optimal f" — the position size that maximises geometric mean growth — and documented that small deviations from optimal sizing produce large deviations in expected outcomes. [Thorp \(1969\)](#) applied the Kelly criterion to trading, showing that position size, not strategy choice, determined whether otherwise-identical strategies produced wealth or ruin. [Tharp \(1998, 2008\)](#) argued from extensive practitioner observation that traders consistently underestimate the importance of sizing and overestimate the importance of entries.

The academic literature on the topic is comparatively thin. [MacLean, Thorp, and Ziemba \(2010\)](#) provide an authoritative reference, but the focus is largely on Kelly-criterion theory rather than on empirical comparison of sizing approaches. [Clenow \(2013, 2015\)](#) applies Monte Carlo methods to demonstrate the dominance of sizing in trend-following systems, and our methodology is closest to his — but with a more systematic variance-decomposition framework and a broader set of sizing approaches.

This note contributes a clean variance-decomposition study: how much of terminal-outcome variance is attributable to position sizing, versus signal selection, versus pure path-dependence noise? The question is empirically tractable through Monte Carlo simulation, and the answer is informative for both system design and end-user education.

The note is structured as follows. Section 2 reviews the theoretical foundations of position sizing. Section 3 describes our Monte Carlo methodology. Section 4 presents the empirical results across sizing approaches. Section 5 examines the variance decomposition. Section 6 discusses implications for design and disclosure.

02 Theoretical foundations

2.1 The Kelly criterion

The Kelly criterion specifies the position size that maximises the long-run geometric growth rate of wealth, given known win probability and payoff distribution. For a binary bet with win probability p , win amount b , and loss of unit stake, the Kelly fraction is $f^* = (bp - q) / b$, where $q = 1 - p$. For continuous return distributions, the analogue is $f^* = \mu / \sigma^2$, where μ is expected excess return and σ^2 is variance of return.

Several properties of the Kelly fraction are important for trading applications:

- **It maximises geometric mean growth, not arithmetic mean.** An investor who cares about compound wealth over long horizons should use Kelly; an investor who cares about expected value at a specific date should use full leverage if expected return is positive.
- **It produces large drawdowns.** Even at full Kelly, the median path experiences 50%+ drawdowns over a long horizon. This is mathematically inherent, not a flaw of the approach.
- **It is sensitive to parameter uncertainty.** If the trader's estimate of expected return or variance is wrong, full Kelly sizing can be dramatically wrong. Practitioners therefore use fractional Kelly (typically 0.25–0.5× full Kelly) to provide a margin of safety against parameter misestimation.

2.2 Volatility targeting

Volatility targeting sizes positions inversely to recent realised volatility, producing constant ex-ante portfolio volatility. [Moreira and Muir \(2017\)](#) show that volatility-managed portfolios produce statistically significant alpha across multiple asset classes; [Harvey, Hoyle, Korgaonkar, Rattray, Sargaison, and Van Hemert \(2018\)](#) extend this to a broader cross-asset framework. The mechanism is that volatility is persistent (high-vol periods tend to follow high-vol periods, per the GARCH literature), so reducing exposure during high-vol periods reduces realised drawdowns without commensurate cost in expected return.

For a trader running a single-strategy system, volatility targeting amounts to scaling position size by $1/\sigma_{\text{recent}}$, where σ_{recent} is the recent realised volatility of the underlying instrument. The approach implicitly increases leverage during low-vol periods (when the strategy is most likely to be profitable per unit of margin) and decreases leverage during high-vol periods (when realised drawdowns are most likely to be large).

2.3 Fixed-fractional and fixed-dollar sizing

Simpler approaches to sizing — fixed-dollar (constant notional per trade) and fixed-fractional (constant percentage of equity per trade) — are common in retail practice but have known weaknesses. Fixed-dollar sizing produces an asymmetric risk profile: drawdowns reduce subsequent buying power proportionally, and recovery requires larger percentage gains to restore initial wealth. Fixed-fractional sizing addresses this by scaling with equity, but does not adjust for changing volatility regimes or for the relationship between position size and expected return.

[Vince \(1990\)](#) argues that fixed-fractional sizing implicitly assumes constant relationship between equity and optimal size, which is rarely true in practice. The approach systematically over-sizes during high-vol regimes (when the same fractional position is more risky in absolute terms) and under-sizes during low-vol regimes.

03 Methodology

3.1 Base strategy

We define a base trend-following strategy with the following specification:

- **Signal:** 12-month price momentum on the underlying instrument. Long when current price is above 12-month moving average, flat otherwise.
- **Universe:** A simulated multi-asset basket modelled on broad-market index, long-duration bonds, and broad commodities, with monthly rebalancing.
- **Calibrated Sharpe:** The strategy is parameterised to produce an in-expectation Sharpe of 0.7 over the simulation period.

3.2 Sizing rules tested

Table 1. Position-sizing rules tested

Sizing rule	Specification
Fixed dollar	Constant \$10K notional per signal, regardless of volatility or equity
Fixed fractional	10% of current equity per signal, regardless of volatility
Volatility-targeted	Position size scaled to produce 10% annualised portfolio volatility
Fractional Kelly (0.25)	25% of full Kelly fraction, rebalanced monthly
Full Kelly	Full Kelly fraction, rebalanced monthly

3.3 Monte Carlo construction

For each of the five sizing rules, we generate 2,000 forty-year equity-curve paths. Each path is constructed by bootstrap-resampling monthly returns from the historical sample, then applying the specified sizing rule with full path-dependent rebalancing. The bootstrap is block-stationary (12-month blocks) to preserve the autocorrelation structure of the underlying signal.

For each path, we record:

- Terminal wealth (equity at end of 40 years, normalised to starting equity = 1.0).
- Maximum peak-to-trough drawdown over the path.
- Geometric mean return (annualised).
- Volatility (annualised standard deviation of monthly returns).
- Realised Sharpe ratio (geometric mean / volatility).
- Indicator for "ruin" (terminal wealth < 25% of starting; equivalent to 75%+ peak-to-trough drawdown without recovery).

Variance decomposition is performed by ANOVA across the 10,000 paths, partitioning total variance in terminal wealth into: (a) between-sizing-rule variance (5 groups), (b) within-sizing-rule variance attributable to bootstrap noise. We separately compute the variance attributable to varying the strategy's signal parameters within each sizing rule.

04 Empirical results

4.1 Cross-sectional outcomes by sizing rule

Table 2. Path outcomes by position-sizing rule (40-year horizon)

Sizing rule	Median wealth	25th %ile	Median MDD	Ruin %
Fixed dollar	2.4×	1.1×	−42%	24%
Fixed fractional	3.1×	1.6×	−38%	18%
Volatility-targeted	5.3×	2.5×	−24%	4%
Fractional Kelly (0.25)	5.1×	2.4×	−27%	5%
Full Kelly	7.2×	1.8×	−51%	31%

VOL-TARGET RUIN %	FULL KELLY RUIN %	FIXED-DOLLAR RUIN %
4%	31%	24%

The cross-sectional results are striking. Volatility-targeted sizing produces both the highest median wealth and the lowest median drawdown across the five rules — a Pareto improvement over the simpler approaches. Fractional Kelly performs nearly as well, with marginally lower median wealth but slightly tighter drawdowns. Both substantially outperform fixed-dollar and fixed-fractional sizing on every metric.

Full Kelly produces the highest median terminal wealth but at the cost of substantial drawdown risk: 31% of paths end in ruin (terminal wealth < 0.25× initial). The ruin probability is concentrated in paths where early adverse runs reduce equity below the threshold from which Kelly sizing can recover within the 40-year horizon. The result is consistent with the theoretical literature — Kelly maximises geometric mean growth in expectation but produces a heavy left tail.

4.2 Variance decomposition

Table 3. Variance decomposition of terminal wealth (10,000 paths)

Source of variance	Share of total
Position sizing rule (across 5 rules)	61.4%
Signal parameters (within sizing rule)	19.7%
Bootstrap path noise	18.9%

Position sizing accounts for approximately 61% of the cross-sectional variance in 40-year terminal wealth — substantially more than the variance attributable to signal-parameter differences across reasonable ranges. The result holds across alternative variance decompositions: when we use maximum drawdown as the dependent variable, position sizing explains approximately 80% of variance; when we use realised Sharpe ratio, approximately 55%.

The implication is clear. A trader who selects a strategy based on signal characteristics alone, without specifying the sizing approach, is determining only ~20% of their eventual outcome distribution. The other 80% is determined by

sizing and pure path-dependence. Sizing, in particular, is the choice the trader actually controls — bootstrap path noise is irreducible.

4.3 The win-rate fallacy

One persistent feature of retail trading literature is the emphasis on "win rate" — the percentage of trades that close profitably — as a primary measure of strategy quality. Our simulations demonstrate the limits of this metric.

Table 4. Outcome distribution by win rate (volatility-targeted sizing)

Win rate	Median wealth	Ruin %
50%	3.8×	7%
55%	4.2×	5%
60%	5.0×	4%
70%	5.6×	3%

With volatility-targeted sizing, win rate has a modest impact on outcomes — moving from 50% to 70% increases median wealth by approximately 50%. With fixed-dollar sizing on the same underlying signals, however, the picture is different:

Table 5. Outcome distribution by win rate (fixed-dollar sizing)

Win rate	Median wealth	Ruin %
50%	1.4×	38%
55%	1.9×	29%
60%	2.4×	24%
70%	3.4×	25%

A 70% win-rate strategy with fixed-dollar sizing still produces ruin in 25% of paths and median wealth less than half what a 50% win-rate strategy with volatility-targeted sizing achieves. The win rate is, by itself, an almost useless summary statistic — it conveys very little about the joint return-risk-sizing package that determines outcomes.

05 Discussion and conclusion

5.1 Why sizing dominates

The mechanical reason that sizing dominates signal selection in our simulations is that sizing rules differ in their treatment of the time-varying volatility of the underlying signal returns. Volatility is persistent (high-vol regimes follow high-vol regimes), and maximum drawdowns occur disproportionately during high-vol regimes. A sizing rule that reduces position size during high-vol regimes (volatility targeting, Kelly) avoids the worst part of the drawdown distribution. A sizing rule that does not (fixed-dollar, fixed-fractional) participates fully in those drawdowns.

The reason sizing matters more than signal-parameter selection within reasonable ranges is that signal parameters in the trend-following family are mostly already correctly chosen — moving from a 10-month to a 14-month lookback window changes the realised Sharpe ratio modestly. Moving from fixed-dollar to volatility-targeted sizing changes the entire shape of the return distribution.

5.2 Implications for system design

Three implications follow from the variance-decomposition results.

System design must include sizing. A "trading system" defined as a set of entry and exit rules without explicit sizing rules is incomplete; it specifies approximately one-fifth of the eventual outcome distribution. Vendors selling trading systems should publish the position-sizing rule alongside the signal rule, with equal prominence.

Backtests using arbitrary sizing are uninformative. A backtest performed with a particular sizing rule does not generalise to other sizing rules. The 10-year track record of a system depends on which sizing rule was used during the backtest; switching sizing rules can produce a 10-year track record with materially different characteristics on the same underlying signal.

End-user education must include sizing. A trader using a system without understanding the sizing rule is operating without understanding the most important determinant of their outcome. The widespread emphasis in retail trading education on entry timing, win rates, and trading psychology — at the expense of sizing — leaves users systematically unprepared for what actually matters.

5.3 Limitations

Our simulations use historical-sample bootstrap, which inherits the historical correlation and volatility structure of the underlying assets. Forward-looking outcomes may differ if the underlying volatility or correlation regime shifts substantially. We test a single strategy family (trend-following); results for other strategy types (mean-reversion, statistical arbitrage, options) may differ in magnitudes but should preserve the qualitative finding. Most importantly, our analysis assumes the trader executes the sizing rule perfectly; in practice, behavioural deviation from the prescribed sizing (typically over-sizing during good periods and under-sizing during drawdowns) is a substantial additional source of poor outcomes that we do not model here.

5.4 Conclusion

Position sizing explains approximately 60% of cross-sectional variance in long-run terminal wealth and approximately 80% of variance in maximum drawdown across our 10,000 simulated paths. The variance attributable to position sizing exceeds the variance attributable to signal-parameter selection by roughly 3:1 within reasonable parameter ranges. The implication is that the conventional emphasis on signal selection — at the expense of sizing — is roughly inverted from what produces good outcomes.

The practical recommendations are straightforward. Volatility-targeted or fractional-Kelly sizing dominates simpler approaches across our simulation universe. Full Kelly produces highest expected geometric mean but unacceptable left-tail outcomes. Fixed-dollar and fixed-fractional sizing produce both lower median outcomes and substantially

worse tail outcomes. The win rate of a strategy is a poor summary statistic; the joint signal-sizing package determines outcomes, and the sizing component is, on average, the larger contributor.

*A 70% win-rate system with poor sizing produces **ruin in 25% of paths**. A 55% win-rate system with proper sizing produces **positive long-run growth in 95% of paths**. The win rate is one of the least informative summary statistics in trading.*

Carl G. C. Eriksson, Bo-Ben Åkesson, Position sizing dominates strategy selection

REF References

- Clenow, A. (2013). *Following the Trend: Diversified Managed Futures Trading*. Wiley.
- Clenow, A. (2015). *Stocks on the Move: Beating the Market with Hedge Fund Momentum Strategies*. Equilateral Publishing.
- Harvey, C. R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M., and Van Hemert, O. (2018). The impact of volatility targeting. *Journal of Portfolio Management*, 45(1): 14–33.
- Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal*, 35(4): 917–926.
- MacLean, L. C., Thorp, E. O., and Ziemba, W. T. (eds.) (2010). *The Kelly Capital Growth Investment Criterion: Theory and Practice*. World Scientific.
- Moreira, A. and Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4): 1611–1644.
- Poundstone, W. (2005). *Fortune's Formula: The Untold Story of the Scientific Betting System That Beat the Casinos and Wall Street*. Hill and Wang.
- Rotando, L. M. and Thorp, E. O. (1992). The Kelly criterion and the stock market. *American Mathematical Monthly*, 99(10): 922–931.
- Tharp, V. K. (2008). *Trade Your Way to Financial Freedom*, 2nd ed. McGraw-Hill.
- Thorp, E. O. (1969). Optimal gambling systems for favorable games. *Review of the International Statistical Institute*, 37(3): 273–293.
- Thorp, E. O. (2006). The Kelly criterion in blackjack, sports betting, and the stock market. In *Handbook of Asset and Liability Management*, vol. 1, 385–428. Elsevier.
- Vince, R. (1990). *Portfolio Management Formulas: Mathematical Trading Methods for the Futures, Options, and Stock Markets*. Wiley.
- Vince, R. (1992). *The Mathematics of Money Management: Risk Analysis Techniques for Traders*. Wiley.

APX **Appendix A: Methodology details**

A.1 Bootstrap construction

Our Monte Carlo paths are constructed using stationary block bootstrap with 12-month blocks. The block length preserves the autocorrelation structure of trend-following returns (which exhibit positive serial correlation at 12-month horizons consistent with the time-series momentum literature). Alternative block lengths (6 months, 18 months, 24 months) produce qualitatively similar results.

For each path, the bootstrap proceeds as follows: (a) randomly select 480 monthly returns from the historical sample with replacement, in 40 blocks of 12 consecutive months each; (b) apply the specified position-sizing rule to produce equity-curve dynamics; (c) record path-level statistics. The 2,000 paths per sizing rule provide sufficient statistical power to detect cross-sectional differences in outcome distributions at conventional significance thresholds.

A.2 Sensitivity to historical sample

We tested using historical samples of varying lengths (50-year, 100-year, full 152-year) and across multiple asset classes (S&P 500, multi-asset basket, futures basket). The qualitative finding — sizing dominates signal selection — is robust across all variants. The quantitative magnitudes shift modestly: longer historical samples include more crisis periods, increasing the spread between vol-targeted and fixed-dollar sizing as the latter is more exposed to the high-vol regimes.

A.3 Practitioner sizing comparison

Table A1. Sizing approach by trader type (industry survey)

Trader type	Modal sizing approach
Retail (uninformed)	Fixed-dollar
Retail (informed)	Fixed-fractional
Professional CTA	Volatility-targeted
Quant hedge fund	Volatility-targeted or fractional Kell

A.4 ProRealTime: volatility-targeted position sizing

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// Volatility-targeted sizing – scale position so realised vol ~ target
// Run on daily bars

DEFPARAM CumulateOrders = False
DEFPARAM PreLoadBars = 250

target_vol_annual = 0.10 // 10% annualised
lookback_vol = 60 // days of realised vol estimation
trading_days = 252

// Realised volatility of daily returns
```

Trader type	Modal sizing approach
	<pre> ret = (close - close[1]) / close[1] realised_vol_daily = STD[lookback_vol](ret) realised_vol_annual = realised_vol_daily * SQRT(trading_days) // Scale factor – how many "units" of the underlying per unit of capital vol_scale = target_vol_annual / realised_vol_annual // Position size in shares/contracts capital_at_risk = strategyprofit + capital notional_target = capital_at_risk * vol_scale shares = notional_target / close // Combined with a momentum signal (example): mom = (close - close[252]) / close[252] IF mom > 0 AND NOT LongOnMarket THEN BUY shares CONTRACT AT MARKET ENDIF IF mom <= 0 AND LongOnMarket THEN SELL AT MARKET ENDIF y </pre>
Statistical arbitrage	Position-by-position Kelly with parameter uncertainty adjustment

The progression from retail to professional sizing approaches mirrors the empirical hierarchy: simpler sizing produces worse expected outcomes, and the professional consensus has converged on volatility-targeting as the dominant approach over the last twenty years.

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